

Exp No:4	Exploring Hugging Face – From Manual ML to Pretrained Models
Date:	

AIM

To explore Hugging Face for model development using both scratch-built and pretrained models.

ALGORITHM

1. Prepare satisfied and unsatisfied feedback datasets.
2. Implement Naïve Bayes from scratch.
3. Calculate word probabilities with Laplace smoothing.
4. Classify test feedback.
5. Load a pretrained Hugging Face sentiment model.
6. Compare both model outputs.

PROGRAM:

Task1:

TrainNaiveBayesFromScratch

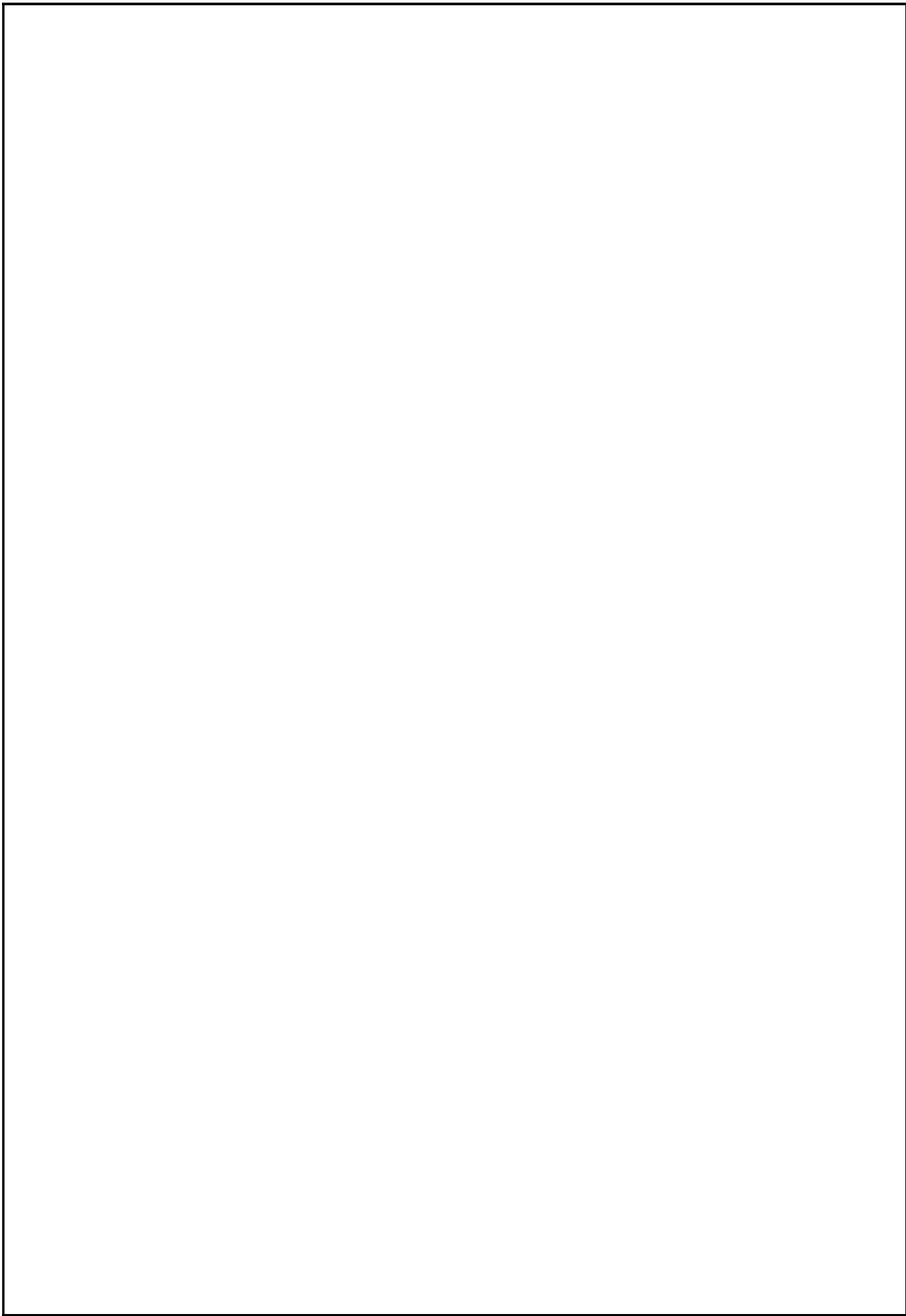
```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
```

```
# Dataset
```

```
messages = [
    "Win a free mobile now",
    "Meeting scheduled at 10 AM",
    "Claim your lottery prize",
    "Project deadline is tomorrow",
    "Limited offer buy now",
    "Team lunch today",
    "Congratulations you won",
    "Please review the document",
    "Free coupons available",
    "Call me when free"
]
```

```
labels = [
    "Spam", "Not Spam", "Spam", "Not Spam", "Spam",
    "Not Spam", "Spam", "Not Spam", "Spam", "Not Spam"
]
```

```
# Convert text to Bag-of-Words
```



```
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(messages)

# Train Naïve Bayes model
model = MultinomialNB()
model.fit(X, labels)

# Test message
test_message = ["Free offer available"]
test_vector = vectorizer.transform(test_message)

prediction = model.predict(test_vector)

print("Test Message:", test_message[0])
print("Predicted Class:", prediction[0])

# Accuracy on training data
train_predictions = model.predict(X)
accuracy = accuracy_score(labels, train_predictions)

print("Model Accuracy:", accuracy)
```

Task2:

PreTrainedModelHuggingFace-Solution

```
from transformers import pipeline
# Load pretrained sentiment analysis model
classifier = pipeline(
    "sentiment-analysis",
    model="distilbert-base-uncased-finetuned-sst-2-english"
)
# Test message
text = "Free offer available"
result = classifier(text)
print("Text:", text)
print("Prediction:", result)
```

OUTPUT:

Task1:

```
Test Message: Free offer available  
Predicted Class: Spam  
Model Accuracy: 1.0  
PS C:\Users\kavim>
```

Task2:

```
Device set to use cpu  
Text: Free offer available  
Prediction: [{'label': 'POSITIVE', 'score': 0.9951098561286926}]  
PS C:\Users\kavim>
```

Preparation	20	
Implementation	30	
Viva	15	
Record	10	
Total	75	

RESULT:

Thus, Hugging Face model development using scratch and pretrained models was successfully explored and uploaded to GitHub.

GitHub Link : <https://github.com/KavimugilRajasekar/MachineLearningLab.git>